

Alexandre Dang

Research Engineer in AI Safety

PhD-level researcher transitioning into AI Safety with cybersecurity, cryptography and verification expertise

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EXPERIENCE

Independent Researcher in AI Safety

January 2026 – Today | Rennes, France

- **SPAR Fellowship** (ongoing since February 2026)
 - Proving model equivalence using challenges and zero-knowledge cryptography
 - Formalized theoretical bounds of inference outputs between a model and its zk-quantized version ([notes](#))
 - Generated proofs of inference of PyTorch models using ezkl and ZKTorch ([repo](#))
 - Implemented transformer block with quantization schemes in Rust ([repo](#))
 - Preparing paper submission to ICML 2026 workshop
- **BlueDot Technical AI project** (ongoing since April 2026)
 - Prevention of code vulnerabilities injection using AI Control
 - Experimentations using ControlArena
 - Incorporating security tools to improve scoring system of untrusted model actions
 - Improving trusted model context with a detailed plan of actions from untrusted model
- Organisation of **public outreach on AI Safety** in Rennes (ongoing since April 2026)
 - Currently meeting with universities of Rennes to organise presentations to students
- Writing a personal literature survey of AI Safety ([notes](#))
- Attended **ML4Good Technical AI bootcamp** (April 2026)
- Attended **BlueDot Technical AI Safety course** (January 2026)

CrabNebula | SOFTWARE CYBERSECURITY ENGINEER

January 2023 – November 2024 | Full remote from Rennes, France

- Lead dev and co-owner of **tauri-fuzz** a Rust-based fuzzer for applications
 - Managed the project from scratch, including design, development and CI
 - Designed a novel approach to test security boundaries of applications
 - **tauri-fuzz** detects vulnerable system calls issued by applications
- Conducted security audits of customers Tauri applications and CrabNebula products
- Software development of CrabNebula products in Rust, Next.js and Typescript

IOTA (blockchain) | RESEARCH SOFTWARE ENGINEER

August 2021 – November 2022 | Full remote from Madrid, Spain

- Rust development of **Stronghold**, a secure engine to store cryptographic secrets
 - Detected core memory vulnerability and applied a fix using the Boojum scheme
 - Researched and coded concurrency in **Stronghold** (resulted in a [blogpost](#))
 - Implemented a new feature for secure key management
- Audited and supported the development of IOTA smart contracts
- Developed a formal model of IOTA distributed ledger in Agda

CentraleSupélec | CYBERSECURITY PROFESSOR

2019 – 2020 | Rennes, France

- Taught Master level courses such as Cryptography, Compilation, Python...
- Created and launched an introductory Rust system programming course

Inria Rennes | PHD ON "SECURE COMPILATION"

2016 – 2019 | Rennes, France

Research: Verified Software Fault Isolation (SFI)

AI SAFETY INTERESTS

zkML
AI Control
Formal verification of AI systems

SKILLS

EXPERTISE

Software security
Formal verification
Cryptography

PROGRAMMING

Python
Rust
C

LANGUAGES

French (Native)
English (Bilingual)
Spanish (Lived 2y in Spain)

EDUCATION

PHD IN COMPUTER SCIENCE

INRIA RENNES
Compilation for Memory Protection
2016-2019, Rennes, France

ENGINEERING DEGREE

CENTRALESUPÉLEC
2016 | Rennes, France

MASTER IN CYBERSECURITY

UNIVERSITY OF RENNES 1
2016 | Rennes, France

RUST BOOTCAMP

LET'S GET RUSTY
2025 (5 months) | Full Remote

- Developed CompCertSFI a C Compiler which produces sandboxed binaries
- Usage of formal methods to prove CompCertSFI safety using Coq/Roq

Research: Preservation of Security during Compilation

- Formalization of security preservation properties for compiler transformations
- Implementation of secure compiler transformation in Coq/Roq
- Benchmarks and optimization of our secure transformations

Publications in international conferences

- [1] Securing Compilation Against Memory Probing, PLAS18
- [2] Compiling Sandboxes: Formally Verified Software Fault Isolation, ESOP19
- [3] Information-Flow Preservation in Compiler Transformations, CSF19

Education: Taught Scala and Python courses at Rennes University

Web development with Ruby on Rails

2015 – 2016 | London, England